

CMT 451 Module 1: Mobile Computing and Threat Landscape. It is designed to be engaging for students, covering the indicative content and mapped standards from your course outline.

CMT 451 Module 1: Mobile Computing & Threat Landscape (10-Slide Summary)

Slide 1: Title & Introduction

CMT 451: Mobile Security

Module 1 – Mobile Computing and Threat Landscape

Instructor: Dr. Stanley Mwangi Chege
(University Name)



Slide 2: Module Overview

Setting the Stage



Understanding Architectures



Identifying Threat Actors



Analyzing Attack Surfaces

Goal: Establish foundational understanding of the environment.

Slide 3: Learning Outcomes

At the end of this module, you will:

- ✓ Identify mobile threats & vulnerabilities.
- ✓ Analyze layers of mobile architecture.
- ✓ Map security challenges to international standards (NIST, ISO)

Slide 4: Mobile Architectures (The Environment)



Complexity: Heterogeneity & Connectivity.

Slide 5: Threat Actors (The Adversaries)



Cybercriminals
(Financial Gain)



Insiders
(Data Leaks)



Nation-States
(Espionage/APTs)



Script Kiddies
(Vandalism)

Who is attacking?

Slide 6: Attack Surfaces (The Entry Points)



Device Layer:
Theft, Juice Jacking



Network Layer:
MitM, Rogue Towers



Application Layer:
Malicious Apps



Ecosystem:
App Stores, Supply Chain

Where are we vulnerable?

Slide 7: Standard Mapping – NIST CSF

NIST
IDENTIFY
Function



ID.AM (Asset Management):
Know your devices



ID.RA (Risk Assessment):
Understand threats & impact

Focus: Identify Assets & Risks.

Slide 8: Standard Mapping – ISO/IEC 27001

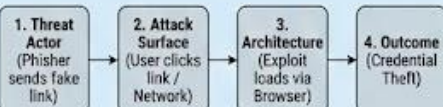
ISO
Annex A
Controls



A.6.2 (Mobile Devices):
Policy & security measures for
BYOD/remote access

Focus: Specific Control Requirements.

Slide 9: Discussion – The "Kill Chain"



Scenario: Phishing SMS (Smishing)

Slide 10: Summary & Next Steps



Architecture is complex & layered



Threat actors vary in motivation



Standards guide defense strategy

Next Module: Mobile OS Security (Android vs. iOS)



Assignment: Review "Threat Landscape" reading.

Slide 1: Title Slide

- **Title:** CMT 451: Mobile Security
 - **Subtitle:** Module 1 – Mobile Computing and Threat Landscape
 - **Instructor:** Dr. Stanley Mwangi Chege
 - **Institution:** CUEA
 - **Visual:**
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Slide 2: Module Overview

- **Theme:** Setting the Stage
- **Key Topics:**
 - Understanding Mobile Architectures
 - Identifying Threat Actors
 - Analyzing Attack Surfaces
- **Goal:** To establish a foundational understanding of the environment we are trying to secure¹.

Speaker Notes: "Welcome to Module 1. Before we can secure devices (Module 2) or apps (Module 3), we must understand the battlefield. Today we look at the 'Threat Landscape'—who is attacking us, where they attack, and the architecture of the systems we defend."

Slide 3: Learning Outcomes (CBET)

- **At the end of this module, you will be able to:**
 1. **Identify** mobile security threats and vulnerabilities unique to mobile computing².
 2. **Analyze** the different layers of mobile architecture.
 3. **Map** security challenges to international standards like NIST CSF and ISO/IEC 27001³.

Speaker Notes: "These outcomes align with the course's main objective: to design and evaluate secure mobile systems using professional frameworks."

Slide 4: Mobile Architectures (The Environment)

- **The Complexity of Mobile:**
 - **Heterogeneity:** Diverse devices, OS versions, and hardware.
 - **The Stack:**

- Hardware (Processors, Sensors, GPS)
- OS Kernel & Middleware
- Application Layer
- **Connectivity:** Cellular (4G/5G), Wi-Fi, Bluetooth, NFC.

Speaker Notes: "Mobile security is harder than desktop security because the architecture is fragmented. We aren't just securing a computer; we are securing a device with GPS, cameras, and constant connectivity to untrusted networks."

Slide 5: Threat Actors (The Adversaries)

- **Who is attacking?** ⁴
 - **Cybercriminals:** Motivated by financial gain (Banking trojans, Ransomware).
 - **Insiders:** Employees (malicious or negligent) leaking data.
 - **Nation-States:** Advanced Persistent Threats (APTs) focused on espionage (e.g., Pegasus spyware).
 - **Script Kiddies:** Using off-the-shelf tools for vandalism.

Speaker Notes: "Understanding the 'Who' helps us predict the 'How'. A cybercriminal wants your credit card; a nation-state wants your location and microphone access."

Slide 6: Attack Surfaces (The Entry Points)

- **Where are we vulnerable?** ⁵
 - **Device Layer:** Physical theft, USB attacks (Juice Jacking).
 - **Network Layer:** Man-in-the-Middle (MitM) attacks over public Wi-Fi; Rogue cell towers (Stingrays).
 - **Application Layer:** Malicious apps, repackaged apps, permission abuse.
 - **Ecosystem:** App stores and supply chain vulnerabilities.

Speaker Notes: "The 'Attack Surface' is the sum of all points where an unauthorized user can try to enter data or extract data. Mobile devices have massive attack surfaces due to their portability and numerous sensors."

Slide 7: Standard Mapping – NIST CSF

- **Framework:** NIST Cybersecurity Framework (CSF)
- **Focus Function: IDENTIFY** ⁶
 - **ID.AM (Asset Management):** Knowing what devices manage your data.

- **ID.RA (Risk Assessment):** Understanding the likelihood and impact of mobile threats.
- **Why it matters:** You cannot protect what you do not know exists.

Speaker Notes: "In this module, we apply the 'Identify' function. We are identifying the assets (Architecture) and the risks (Threat Actors)."

Slide 8: Standard Mapping – ISO/IEC 27001

- **Framework:** ISO/IEC 27001 Information Security Management
- **Focus:** Annex A Controls ⁷
 - **A.6.2 (Mobile Devices and Teleworking):** Policy and security measures to protect information accessed, processed, or stored on mobile devices.
 - **Relevance:** Corporate governance requires specific policies for BYOD (Bring Your Own Device) and remote access.

Speaker Notes: "While NIST gives us the strategy (Identify), ISO 27001 Annex A gives us the specific control requirement: You *must* have a policy for mobile devices."

Slide 9: Discussion – The "Kill Chain"

- **Scenario:** A Phishing SMS (Smishing).
 1. **Threat Actor:** Cybercriminal sending a fake bank link.
 2. **Attack Surface:** The user (Social Engineering) and the Network.
 3. **Architecture:** Exploit loads via the web browser (Application Layer).
 4. **Outcome:** Credential Theft.
- **Question:** Which layer of the architecture failed? (User? OS? Network?)

Speaker Notes: "Let's apply what we learned. If a user clicks a bad link, the architecture did exactly what it was designed to do—open a webpage. This highlights why 'User Awareness' is part of the defense."

Slide 10: Summary & Next Steps

- **Key Takeaways:**
 - Mobile architecture is layered and complex.
 - Threat actors vary in motivation and capability.
 - Standards like NIST (Identify) and ISO 27001 guide our defense strategy.
- **Next Module:** Mobile Operating Systems Security (Android vs. iOS)⁸.

- **Assignment:** Review the "Threat Landscape" section of the reading list.

Quiz: CMT 451 Module 1 – Mobile Computing and Threat Landscape

1. What is the primary title of Module 1 in the CMT 451 course outline?
 - A. Mobile Application Security
 - B. Mobile Operating Systems Security
 - C. Mobile Computing and Threat Landscape
 - D. Emerging Trends and Secure Mobile Design
2. Which of the following is explicitly listed as "Indicative Content" for Module 1?
 - A. Android security
 - B. Mobile architectures
 - C. Secure coding
 - D. Reverse engineering
3. Which NIST Cybersecurity Framework (CSF) function is mapped to Module 1?
 - A. Protect
 - B. Detect
 - C. Respond
 - D. Identify
4. According to the syllabus, which ISO standard is referenced in Module 1?
 - A. ISO/IEC 27001 Annex A
 - B. ISO 22301
 - C. ISO 9001
 - D. ISO 31000
5. Which of the following topics is NOT covered in Module 1?
 - A. Threat actors
 - B. Attack surfaces
 - C. iOS security
 - D. Mobile architectures
6. The "Identify" function of the NIST CSF is relevant to Module 1 because this module focuses on:
 - A. Detecting malware after it enters the system

- B. Understanding the environment and potential risks (Threat Landscape)
 - C. Implementing specific controls for Android devices
 - D. responding to incidents using forensics
7. Which course objective is most directly aligned with the content of Module 1?
- A. Design and evaluate secure mobile systems
 - B. Identify mobile security threats and vulnerabilities
 - C. Apply international cybersecurity standards to mobile environments
 - D. Conduct mobile forensics
8. In the context of the "Threat Landscape" covered in Module 1, "Attack surfaces" generally refer to:
- A. The physical screen of the mobile device
 - B. The specific individuals launching cyberattacks
 - C. The points in the system where an unauthorized user can try to enter or extract data
 - D. The security policies written by the IT department
9. Module 1 serves as the foundation for the course by introducing:
- A. The specific coding languages used for apps
 - B. The high-level architecture and the adversaries (threat actors)
 - C. The detailed configuration of Mobile Device Management (MDM)
 - D. The legal requirements for privacy
10. Which Module 1 topic would involve analyzing the motivation and capabilities of cybercriminals or nation-states?
- A. Mobile architectures
 - B. Attack surfaces
 - C. Threat actors
 - D. Mapped Standards

Answer Key and Explanations

1. **C** – The title of Module 1 is explicitly listed as "Mobile Computing and Threat Landscape"¹.
2. **B** – "Mobile architectures" is listed under the Indicative Content for Module 1². (Android security is Module 2³, Secure coding is Module 3⁴, Reverse engineering is Module 4⁵).
3. **D** – The syllabus maps "NIST CSF - Identify" to Module 1

4. **A** – "ISO/IEC 27001 Annex A" is the mapped standard for Module 1⁷.
5. **C** – "iOS security" is a topic covered in **Module 2** (Mobile Operating Systems Security), not Module 1⁸.
6. **B** – The NIST function "Identify" aligns with the module's focus on understanding "Mobile architectures," "Threat actors," and "Attack surfaces"
7. **B** – The content of Module 1 (Threat Landscape) directly supports the course objective to "Identify mobile security threats and vulnerabilities"¹⁰.
8. **C** – "Attack surfaces" is a key indicative content topic in Module 1 that defines the exposure points of the system¹¹.
9. **B** – Module 1 covers "Mobile architectures" and "Threat actors"¹², setting the stage before specific controls are discussed in later modules.
10. **C** – The topic "Threat actors" deals with the entities behind the attacks¹³.